Worker Quality, Matching and Productivity Slowdown

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Outline

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Views and errors are solely those of the authors

Introduction

Background

- The slowdown of average labor productivity since 2000 is mostly attributed to total factor productivity (Data)
- The slowdown of total factor productivity since 2000 is accompanied with declines in the income share by top earners (Data)
- The income share by bottom 50% earners after 2000 tilted slightly (Data)

This paper

To what extent top workers and their matching with other workers may have contributed to the slowdown of measured productivity

- Did firms choose quantity over quality of the work force?
- Quantitatively, importance of top-quality worker in firm productivity
- Matching efficiency and productivity

We estimate production function incorporating worker matching

Data: Matched employer-employee data in Canada from 2001 to 2015

Preview of main findings

- If distributions of worker quality are Pareto
  - Positive assortative matching between top worker and the rest
  - Matching function is linear in worker quality
- Declines in worker quality explains declines in measured total factor productivity over 2003-2015
  - While the Hicks-neutral technology index increased
- Reallocation fully accounts for declines of measured productivity
  - Unweighted average of firm-level measured productivity rose
- Match efficiency declined from 2003 to 2015

Literature

- Assortative matching with endogenous firm size
    - CES-type: Eeckhout and Kircher (2018)
    - Cobb-Douglas: Grossman, Helpman and Kircher (2017)
- Effect of re-organizing management structure on productivity
    - Hierarchy of work force: Caliendo et al. (2020), Garicano (2000)
- Effects of management practices on productivity
    - Bender etal (2018)

Production and Assignment

Production and worker matching

Production consists of a team of one y-type worker and $l(x)$ workers of type x

$$f(\omega, x, y, l) = e^{\omega}\left[\alpha_x (e^{\omega_x} x)^{1-\sigma} + \alpha_y y^{1-\sigma}\right]^{\frac{\theta}{1-\sigma}} \cdot [l(x)]^{\alpha_l}$$

- ω is the (logarithm of) Hicks-neutral technology, exogenous
- ω_x measures match efficiency, exogenous

The firm's choice

The type-ω firm hires a team to maximize profits

$$\max_{\{x,y,l(x)\}} f(\omega, x, y, l(x)) - w(x)l(x) - r(y)$$

The first-order necessary conditions (FONCs) with respect to y, x, and $l(x)$

$$f_y - r'(y) = 0; \quad f_x - w'(x)l(x) = 0; \quad f_l - w(x) = 0$$

Let the matching function be $y = T(x)$

Let h be the pdf of y, g be the pdf of x

- If positive assortative matching (PAM):

$$\int_{T(x)}^{\bar{y}} l(s)h(s)ds = \int_{x}^{\bar{x}} g(s)ds$$

- If negative assortative matching (NAM):

$$\int_{T(x)}^{\bar{y}} l(s)h(s)ds = \int_{\bar{x}}^{x} g(s)ds$$

Equilibrium is defined by Firm's FONCs + market clearing condition

Equilibrium consists of differential equations

Under PAM

$$f_x - w'(x)l(x) = 0; \quad f_l - w(x) = 0; \quad T'(x) = \frac{\mathcal{H}(x)}{l(x)}$$

- $\mathcal{H}(x) = g(x)/h(T(x))$
- $T'(x)$ is obtained from the market clearing condition

Under NAM

$$f_x - w'(x)l(x) = 0; \quad f_l - w(x) = 0; \quad T'(x) = -\frac{\mathcal{H}(x)}{l(x)}$$

Focus on PAM: it requires $(\theta > 0, \sigma \geq 1)$ or $(\theta < 0, \sigma \leq 1)$

Consider equilibrium matching given y

Equilibrium matching with Pareto distributions

Pareto distributions

- CDF: $G(x) = 1 - P(X > x) = 1 - (\underline{x}/x)^{\lambda_x}$; PDF: $g(x) = \lambda_x \underline{x}^{\lambda_x}/x^{\lambda_x+1}$
- CDF: $H(y) = 1 - P(Y > y) = 1 - \left(\underline{y}/y\right)^{\lambda_y}$; PDF: $h(y) = \lambda_y \underline{y}^{\lambda_y}/y^{\lambda_y+1}$

Then optimal matching is given by

$$y = T(x) = \Psi^{\frac{1}{1-\sigma}} e^{\omega_x} x,$$

with $\Psi = \frac{\alpha_x(1-\alpha_l)[\theta+\alpha_1(\lambda_y-\lambda_x)]}{\alpha_y\alpha_l[\theta-(1-\alpha_l)(\lambda_y+\lambda_x)]}$

The optimal choice on the number of workers

$$l(x) = \Psi^{\frac{\lambda_y}{1-\sigma}} C \cdot e^{\omega_x \lambda_y} x^{\lambda_y - \lambda_x}$$

Optimal wage of x-type workers

$$w(x) = \Lambda \cdot e^{\omega} \cdot e^{\omega_x(\theta - \lambda_y(1-\alpha_1))} \cdot x^{\theta + (1-\alpha_l)(\lambda_x - \lambda_y)}$$

Estimation

Production function under optimal matching

Taking into account the optimal matching, the production function becomes

$$f(\omega, x, T(x), l) = (\alpha_x + \alpha_y \Psi)^{\frac{\theta}{1-\sigma}} e^{\omega} (e^{\omega_x} x)^{\theta} [l(x)]^{\alpha_l}$$

Alternatively,

$$f\left(\omega, T^{-1}(y), y, l\right) = \left(\alpha_x \Psi^{-1} + \alpha_y\right)^{\frac{\theta}{1-\sigma}} e^{\omega} y^{\theta} \left[l\left(T^{-1}(y)\right)\right]^{\alpha_l}$$

- Use this function for estimation

Data source

Canadian Employer Employee Dynamics Database (CEEDD) 2001-2015

- T1 personal tax file: age, sex, marital status, incomes
- T4–ROE: firm-worker matches, annual earnings on each job
- T2S50: business ownership
- NALMF: firm income statement, balance sheets

Measuring worker quality

Abowd etal (1999): two-way fixed effects

$$\ln w_{it} = \beta_0 + \alpha_i + X_{it}\beta + \psi_{j(i,t)} + \varepsilon_{ijt}$$

- α_i is the worker fixed effect
- $\psi_{j(i,t)}$ is the firm fixed effect
- X_{it} includes quadratic and cubic of worker age interacted with gender

Data process:

- Include two-year dummy variables
- Use data sample 2003-2015, age 20 to 64
- Drop matches if earnings are below minimum wage times 40 times 13
- Exclude workers who own the firm

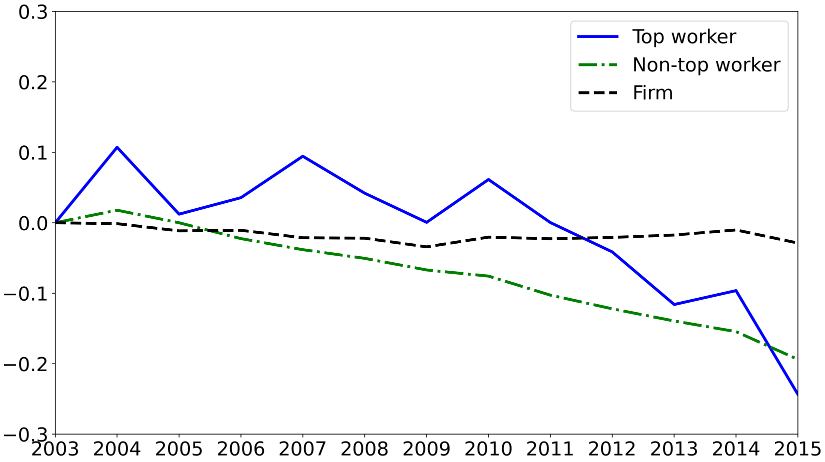


Figure 1: Aggregate fixed effects

Label $h_{it} = \widehat{\alpha}_i + X_{it}\widehat{\beta}$ as quality of worker i

- Top-worker quality: $y = h_{i't}$
  - i' is the highest-paid worker in firm j in year t
- Other-worker quality: $x = \frac{1}{n_{jt}-1} \sum_{i \neq i'} h_{it}$
  - n_{jt} is the employment size of firm j

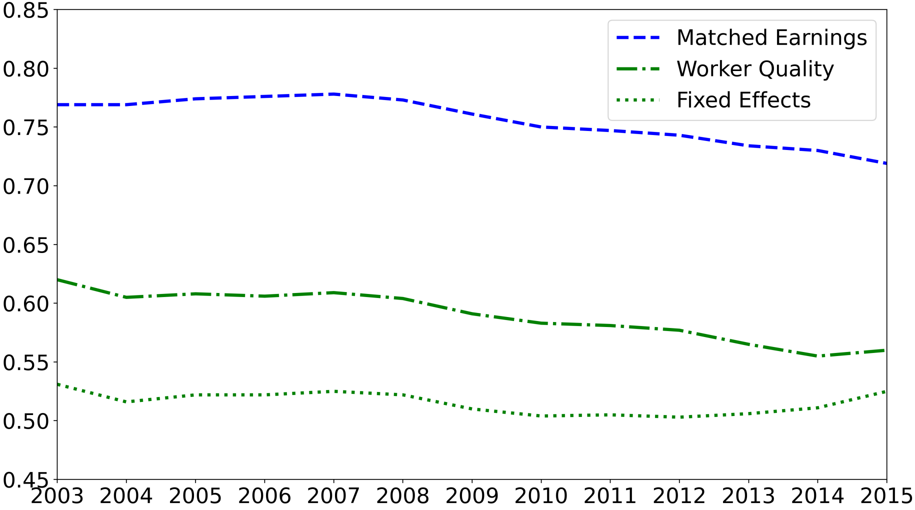


Figure 2: Median log-difference, top workers over non-top workers

Estimating the match efficiency

- Use the match function: $\ln y_{jt} = b_0 + \omega_{xjt} + \ln x_{jt}$
- Assume an AR(1) process: $\omega_{xjt} = \rho_x \omega_{xjt-1} + u_{jt}$ with $u_{jt} \backsim N(0, \sigma_u^2)$
- Estimation by the Generalized Method of Moments (GMM)

$$(\ln y_{jt} - \ln x_{jt}) = (1 - \rho_x) b_0 + \rho_x (\ln y_{jt-1} - \ln x_{jt-1}) + u_{jt}$$

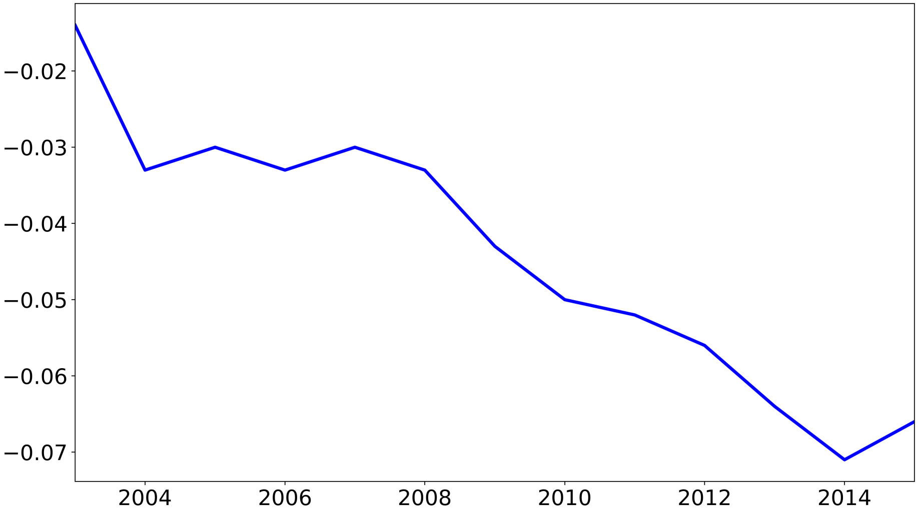


Figure 3: Median of match efficiency ω_x

Production function estimation

Augment production function to include capital stock k_{jt}

The production function of firm j in year t is

$$f\left(T^{-1}(y_{jt}), y_{jt}, l_{jt}, k_{jt}\right) = \left(\alpha_x \Psi^{-1} + \alpha_y\right)^{\frac{\theta}{1-\sigma}} e^{\omega_{jt}} y_{jt}^{\theta} \left[l\left(T^{-1}(y_{jt})\right)\right]^{\alpha_l} k_{jt}^{\alpha_k}$$

In natural logarithm,

$$\ln f_{jt} = \beta_0 + \theta \ln y_{jt} + \alpha_l \ln l_{jt} + \alpha_k \ln k_{jt} + \omega_{jt} + \varepsilon_{jt}$$

- ω_{jt} follows an AR(1) process
- ϵ_{jt} can be the measurement error

Estimation by 2-digit NAICs sector, following Ackerberg, Caves and Frazer (2015) (Details)

Aggregate Productivity

Measured total factor productivity

Recover Hicks-neutral technology index

$$\omega_{jt} = \widehat{\phi}_{jt} - \widehat{\beta}_0 - \widehat{\theta} \ln y_{jt} - \widehat{\alpha}_l \ln l_{jt} - \widehat{\alpha}_k \ln k_{jt}$$

- $\widehat{\phi}_{jt}$: output explained by production function, $\ln f_{jt} = \widehat{\phi}_{jt} + \varepsilon_{jt}$
- Note, coefficient estimates are sector specific

Measured firm-level total factor productivity

$$z_{jt} = \omega_{jt} + \widehat{\theta} \ln y_{jt}, \text{ or}$$

$$z_{jt} = \omega_{jt} + \widehat{\theta}\omega_{xjt} + \widehat{\theta} \ln x_{jt}$$

Measured total factor productivity has two components

- Hicks-neutral technology; Worker quality

Aggregate total factor productivity

Let s_{jt} be firm j's share of output (value added) in aggregate output

Measured total factor productivity (in logarithm)

$$z_t = \omega_t + \tilde{y}_t$$

- $\omega_t = \sum_j^{n_t} s_{jt}\omega_{jt}$
- $\tilde{y}_t = \sum_j^{n_t} s_{jt}\widehat{\theta} \ln y_{jt}$

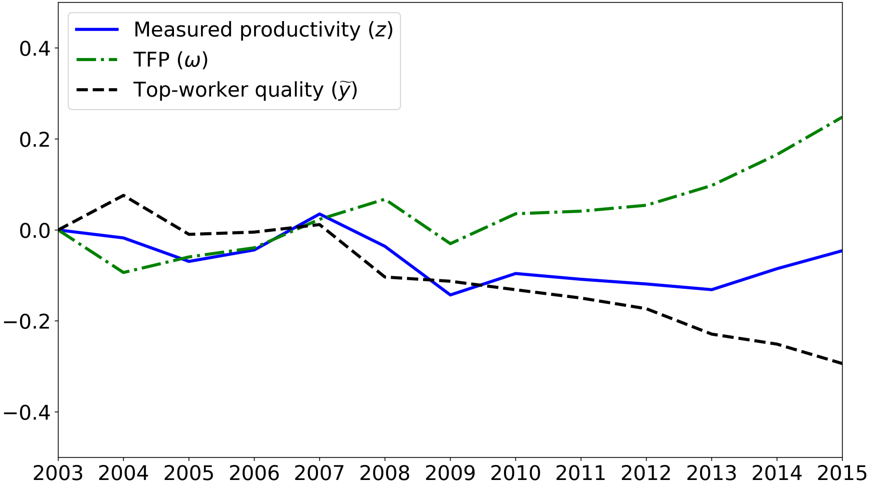


Figure 4: Aggregate productivity measures

Reallocation accounts for productivity slowdown

Following Olley and Pakes (1996),

$$z_t = \bar{z}_t + \sum_{j=1}^{n_t} (s_{jt} - \bar{s}_t)(z_{jt} - \bar{z}_t)$$

- $\bar{z}_t$ is unweighted mean over all firms

Further decomposition

$$z_t = \overline{\omega}_t + \sum_{j=1}^{n_t} (s_{jt} - \bar{s}_t)(\omega_{jt} - \overline{\omega}_t) + \bar{\tilde{y}}_t + \sum_{j=1}^{n_t} (s_{jt} - \bar{s}_t)(\tilde{y}_{jt} - \bar{\tilde{y}}_t)$$

Table 1: Average annual growth rates (percent)

	2003-2015	2003-2008	2008-2015
Aggregate Measured Productivity	-0.38	-0.72	-0.14
Unweighted average	0.61	0.65	0.58
Covariance	-0.99	-1.37	-0.72
Aggregate TFP	2.07	1.35	2.58
Unweighted average	1.58	1.47	1.66
Covariance	0.49	-0.12	0.92
Aggregate contribution of top worker	-2.45	-2.07	-2.71
Unweighted average	-0.97	-0.82	-1.07
Covariance	-1.48	-1.24	-1.64

Top-worker quality over time

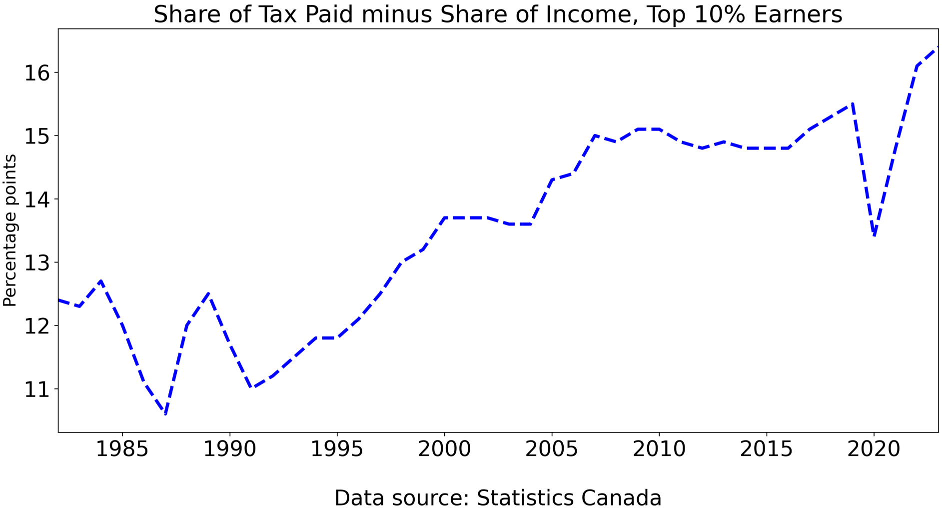
Apply the Olley-Pakes decomposition to top-worker quality

Table 2: Aggregate top-worker quality, average annual growth rates (percent)

Period	2003-2015	2003-2008	2008-2015
Aggregate top-worker quality	-2.60	0.55	-4.85
Unweighted average	-1.56	-1.26	-1.77
Covariance	-1.04	1.81	-3.08
Aggregate top-worker fixed effect	-2.03	0.84	-4.08
Unweighted average	-1.43	-1.38	-1.47
Covariance	-0.60	2.22	-2.61

Why did worker quality decline?

- Brain drain: emigration of top talents?
  - MacGee and Rodrigue (2025): Top earners flowing from CAN to USA increased since 2001
- Taxation on top earners?
  - Share of tax paid relative to share of income has been rising for top 10% earners
- Could increases in tax be a cause of emigration?


Share of Tax Paid minus Share of Income, Top 10% Earners
Percentage points
16
15
14
13
12
11
1985
1990
1995
2000
2005
2010
2015
2020
Data source: Statistics Canada

Conclusion

- Slowdown of measured productivity since 2000 is attributed to
    - Reallocation
    - Declines in measured worker quality
- Top-worker quality declined
    - Declined match efficiency
    - Declined non-top worker quality
- Why did worker quality decline?

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Additional Slides

Growth Accounting of Average Labor Productivity

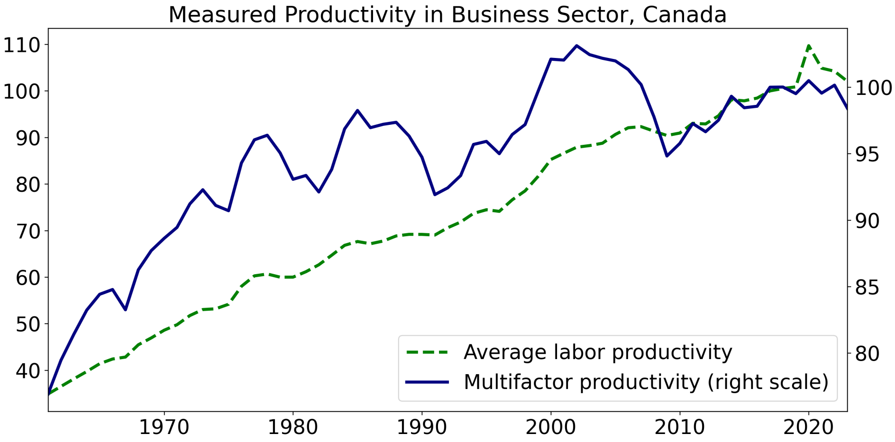
Table 3: Average annual growth rates (percent), business sector

	1961 - 2023	1982 - 2000	2000 - 2019	2019 - 2023
Average Labor Productivity:	1.73	1.75	1.06	0.30
- *K*-Hrs Ratio, contribution	0.93	0.87	0.77	0.40
- *L*-Hrs Ratio, contribution	0.38	0.38	0.27	0.29
- Total Factor Productivity	0.42	0.50	0.02	-0.39

Data source: Statistics Canada Table 36100208

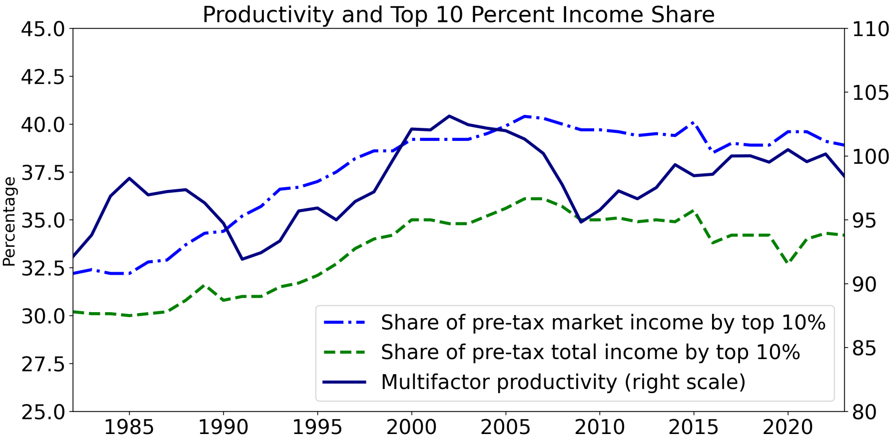
- Total factor productivity differs from Statistics Canada's calculation
- Because here the factor shares use two-year averages (Back)

Measured Productivity in Canada



Data source: Statistics Canada

Productivity and Top 10%



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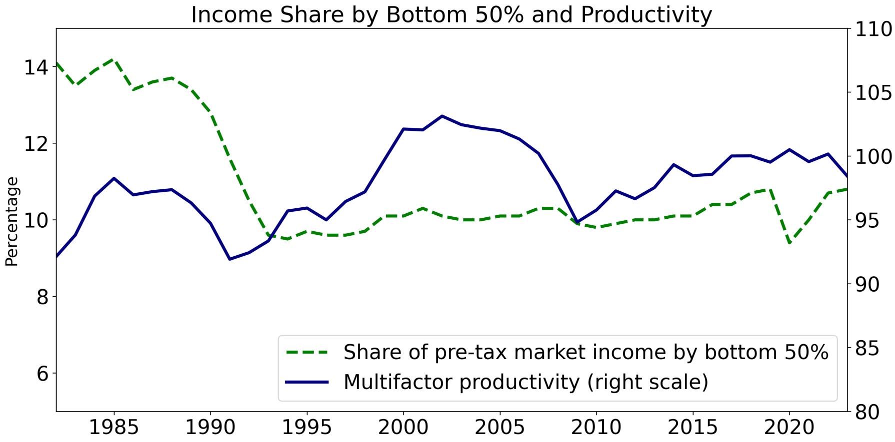
Productivity and Bottom 50%



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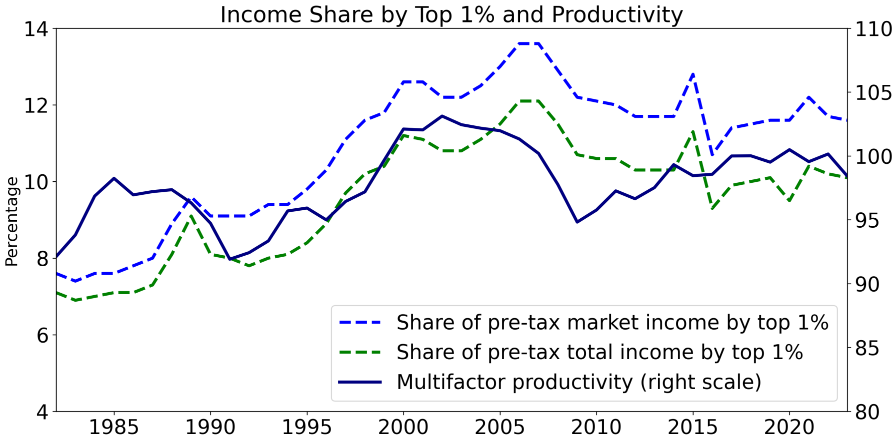
Productivity and Top 1%



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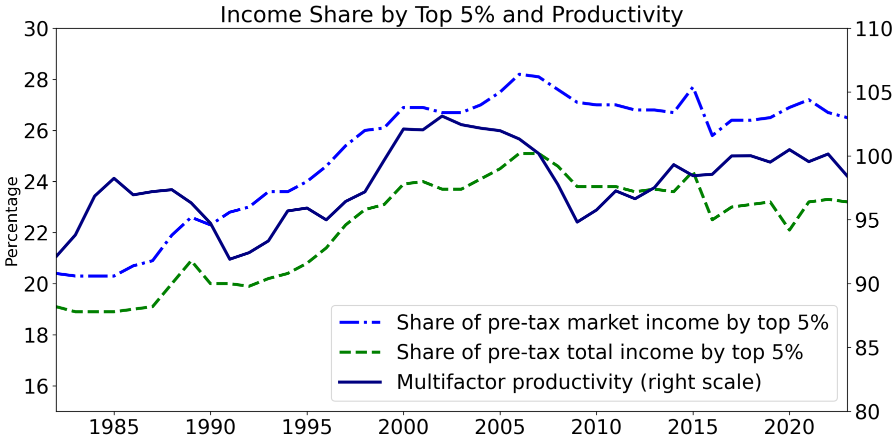
Productivity and Top 5%



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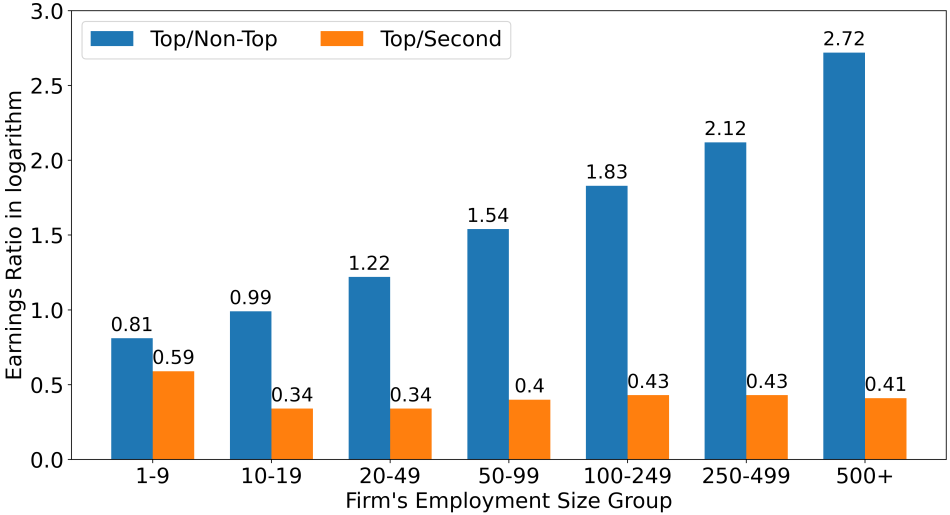

3.0
2.5
2.0
1.5
1.0
0.5
0.0
Earnings Ratio in logarithm
Top/Non-Top
Top/Second
0.81
0.59
0.99
0.34
1.22
0.34
1.54
0.4
1.83
0.43
2.12
0.43
2.72
0.41
1-9
10-19
20-49
50-99
100-249
250-499
500+
Firm's Employment Size Group

Correlation at the firm level

Size	all	1 - 9	10 - 19	20 - 99	100 - 499	500+
# firm/year	6.66m	4.83m	0.91m	0.76m	0.12m	29085
Var(firm FE)	0.12	0.13	0.07	0.06	0.06	0.06
Var(top FE)	0.50	0.36	0.36	0.47	0.58	0.85
Var(non-top FE)	0.20	0.22	0.11	0.10	0.08	0.07
Corr(firmFE,topFE)	0.12	-0.04	0.26	0.29	0.17	0.01
Corr(firmFE,nontopFE)	0.04	-0.08	0.36	0.49	0.54	0.38
Corr(topFE,nontopFE)	0.60	0.57	0.63	0.60	0.49	0.32

Turnover of top-paid workers

At the firm level

- About 26% of firm/year observations saw new top-paid workers
- This turnover rate is higher for lager firms

At the worker level

- 29% of top-worker/year is new, of which
  - Slightly more than half within firm, the rest joining from other firms
- 25% of top-worker/year no longer top next year, of which
  - 56% become non-top worker in the same firm
  - the rest joining other firms as non-top workers
- 6% top-worker/year move to another firm as top worker

Productivity dispersion

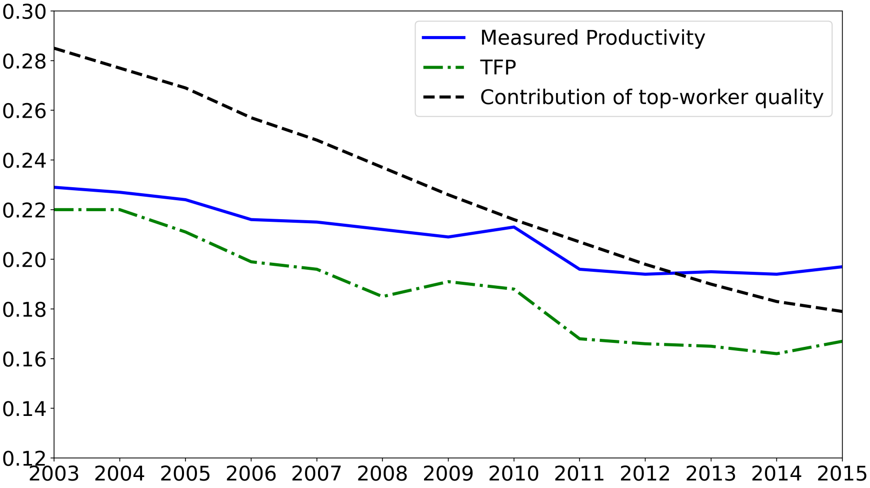


Figure 5: Variance of aggregate measures

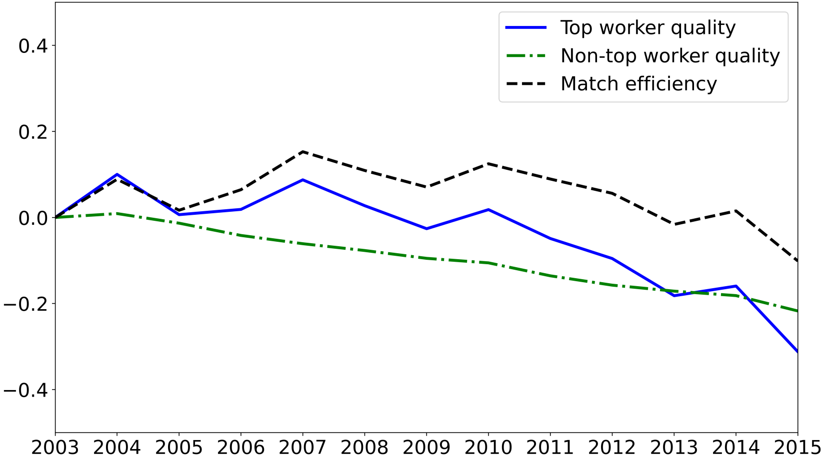
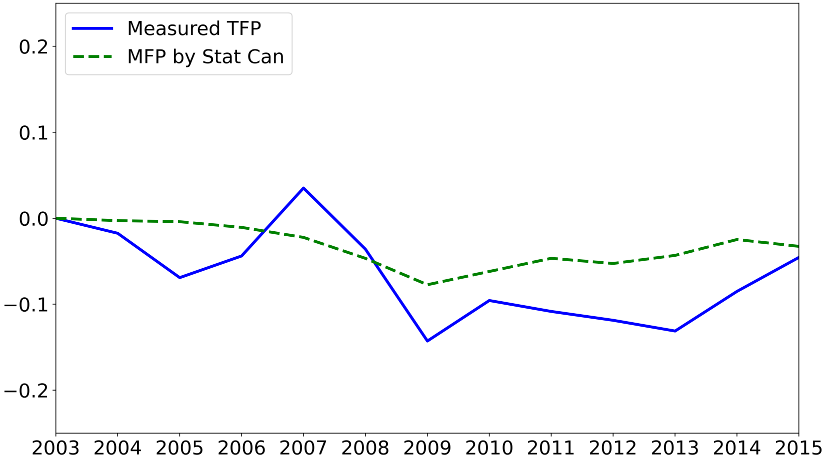
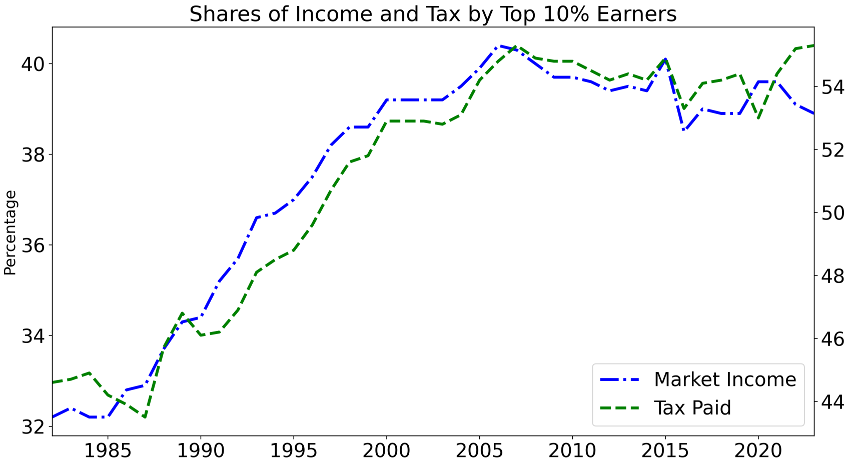


Figure 6: Worker Quality and Match Efficiency, Aggregate

Comparison with Statistics Canada's calculation



Income and Tax, by top 10% earners



Production function estimation details

- Use intermediate inputs as proxy variable, $m_{jt} = M(\omega_{jt}, y_{jt}, l_{jt}, k_{jt}, p_{gt}, p_{mt})$
  - p_{gt} and p_{mt} are respectively the prices of gross output and intermediate inputs, both at 3-digit NAICS
- The inverse of proxy variable is $\omega_{jt} = M^{-1}(m_{jt}, y_{jt}, l_{jt}, k_{jt}, p_{gt}, p_{mt})$. Substituting it for ω_{jt}, the production function becomes

$$\ln f_{jt} = \phi(m_{jt}, y_{jt}, l_{jt}, k_{jt}, p_{gt}, p_{mt}) + \varepsilon_{jt}, \tag{1}$$

- $\phi(m_{jt}, y_{jt}, l_{jt}, k_{jt}, p_{gt}, p_{mt}) =$ $\beta_0 + \theta \ln y_{jt} + \alpha_l \ln l_{jt} + \alpha_k \ln k_{jt} + M^{-1}(m_{jt}, y_{jt}, l_{jt}, k_{jt}, p_{gt}, p_{mt})$

(Back)

- Taken into account $\omega_{jt} = \rho\omega_{jt-1} + \xi_{jt}$, the production function becomes

$$\begin{aligned} \ln f_{jt} = \beta_0 + \theta \ln y_{jt} + \alpha_l \ln l_{jt} + \alpha_k \ln k_{jt} \\ + \rho \left[\phi_{jt-1} - (\beta_0 + \theta \ln y_{jt-1} + \alpha_l \ln l_{jt-1} + \alpha_k \ln k_{jt-1})\right] + \xi_{jt} + \varepsilon_{jt} \end{aligned} \tag{2}$$

The first stage, estimate Eq. (1) by the ordinary least squares (OLS)

- Approximate ϕ_{jt} with a 3rd-order polynomial of $\ln y_{jt}$, $\ln l_{jt}$, $\ln k_{jt}$, $\ln m_{jt}$ and the logarithm of prices
- Obtain the estimate $\widehat{\phi}(m_{jt}, y_{jt}, l_{jt}, k_{jt}, p_{gt}, p_{mt})$

The second stage, substitute $\widehat{\phi}_{jt-1}$ for ϕ_{jt-1} in Eq. (2), and estimate it by GMM

- Use $\mathbf{I}_{jt-1} = (1, \widehat{\phi}_{jt-1}, \ln m_{jt-1}, \ln y_{jt-1}, \ln l_{jt-1}, \ln k_{jt})$ as instrument variables
- The moment conditions are given by

$$E\left[(\xi_{jt} + \varepsilon_{jt}) \otimes \mathbf{I}_{jt-1}\right] = 0$$